

## Toxic Threat Zone

ALOHA® 5.4.7



Time: July 20, 2025 0145 hours PDT (using computer's clock)

Chemical Name: PROPANE

Wind: 1.55 meters/second from 271° true at 10 meters

### THREAT ZONE:

Model Run: Heavy Gas

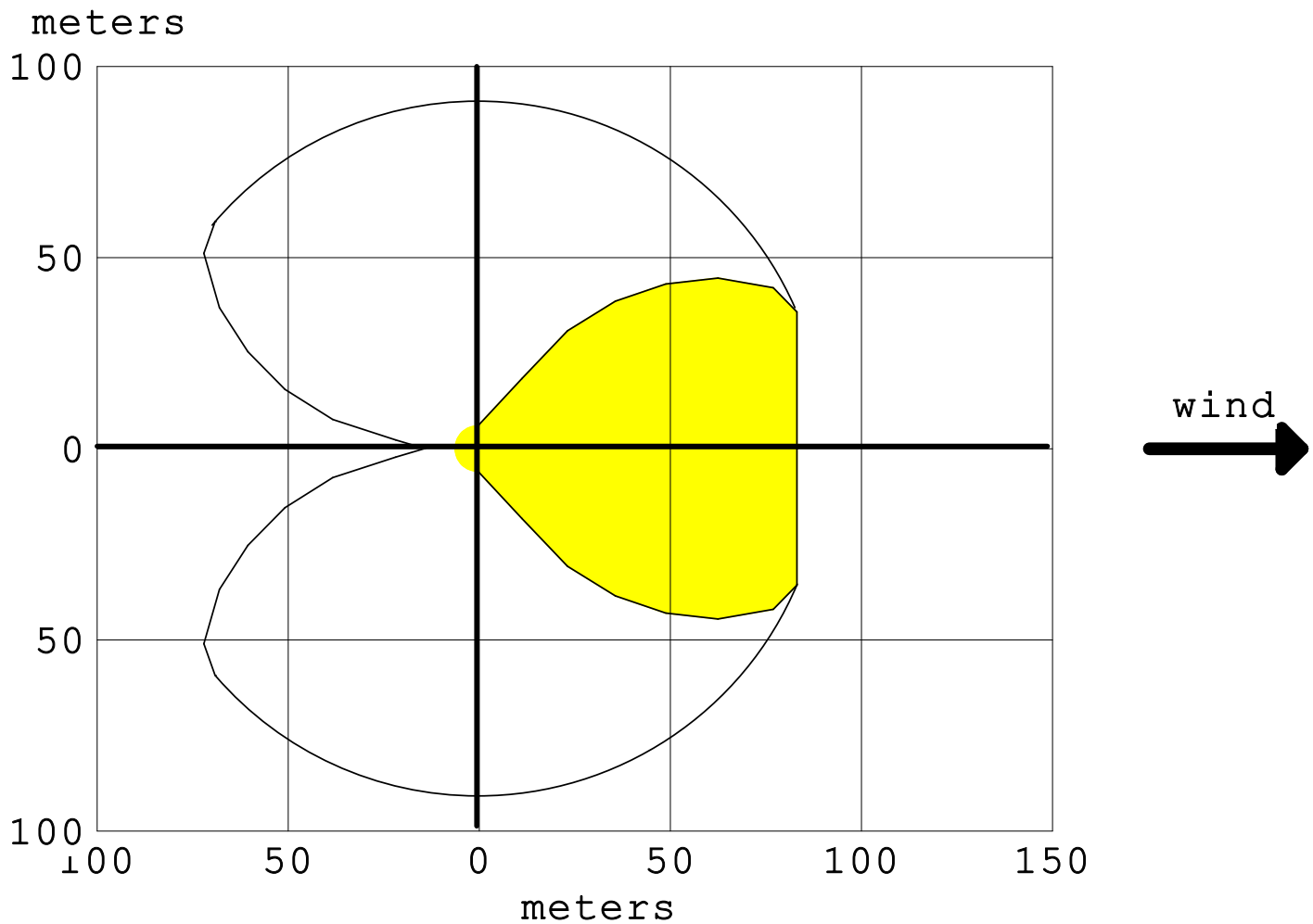
Red : 34 meters --- (33000 ppm = AEGL-3 [60 min])

Note: Threat zone was not drawn because effects of near-field patchiness make dispersion predictions less reliable for short distances.

Orange: 47 meters --- (17000 ppm = AEGL-2 [60 min])

Note: Threat zone was not drawn because effects of near-field patchiness make dispersion predictions less reliable for short distances.

Yellow: 84 meters --- (5500 ppm = AEGL-1 [60 min])



greater than 33000 ppm (AEGL-3 [60 min]) (not drawn)



greater than 17000 ppm (AEGL-2 [60 min]) (not drawn)



greater than 5500 ppm (AEGL-1 [60 min])



wind direction confidence lines

## Text Summary

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### SITE DATA:

Location: TULARE, CALIFORNIA, USA, CALIFORNIA  
Building Air Exchanges Per Hour: 0.28 (unsheltered single storied)  
Time: July 20, 2025 0145 hours PDT (using computer's clock)

### CHEMICAL DATA:

Chemical Name: PROPANE  
CAS Number: 74-98-6 Molecular Weight: 44.10 g/mol  
AEGL-1 (60 min): 5500 ppm AEGL-2 (60 min): 17000 ppm AEGL-3 (60 min):  
33000 ppm  
IDLH: 2100 ppm LEL: 21000 ppm UEL: 95000 ppm  
Ambient Boiling Point: -42.3° C  
Vapor Pressure at Ambient Temperature: greater than 1 atm  
Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

### ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)

Wind: 1.55 meters/second from 271° true at 10 meters  
Ground Roughness: open country Cloud Cover: 4 tenths  
Air Temperature: 18.54° C  
Stability Class: D (user override)  
No Inversion Height Relative Humidity: 54%

### SOURCE STRENGTH:

Leak from hole in vertical cylindrical tank  
Flammable chemical escaping from tank (not burning)  
Tank Diameter: 5.8 meters Tank Length: 12.9 meters  
Tank Volume: 341 cubic meters  
Tank contains liquid Internal Temperature: 18.54° C  
Chemical Mass in Tank: 146,073 kilograms  
Tank is 85% full  
Circular Opening Diameter: 1 centimeters  
Opening is 12.62 meters from tank bottom  
Release Duration: ALOHA limited the duration to 1 hour  
Max Average Sustained Release Rate: 70.7 kilograms/min  
(averaged over a minute or more)  
Total Amount Released: 4,217 kilograms  
Note: The chemical escaped as a mixture of gas and aerosol (two phase flow).

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